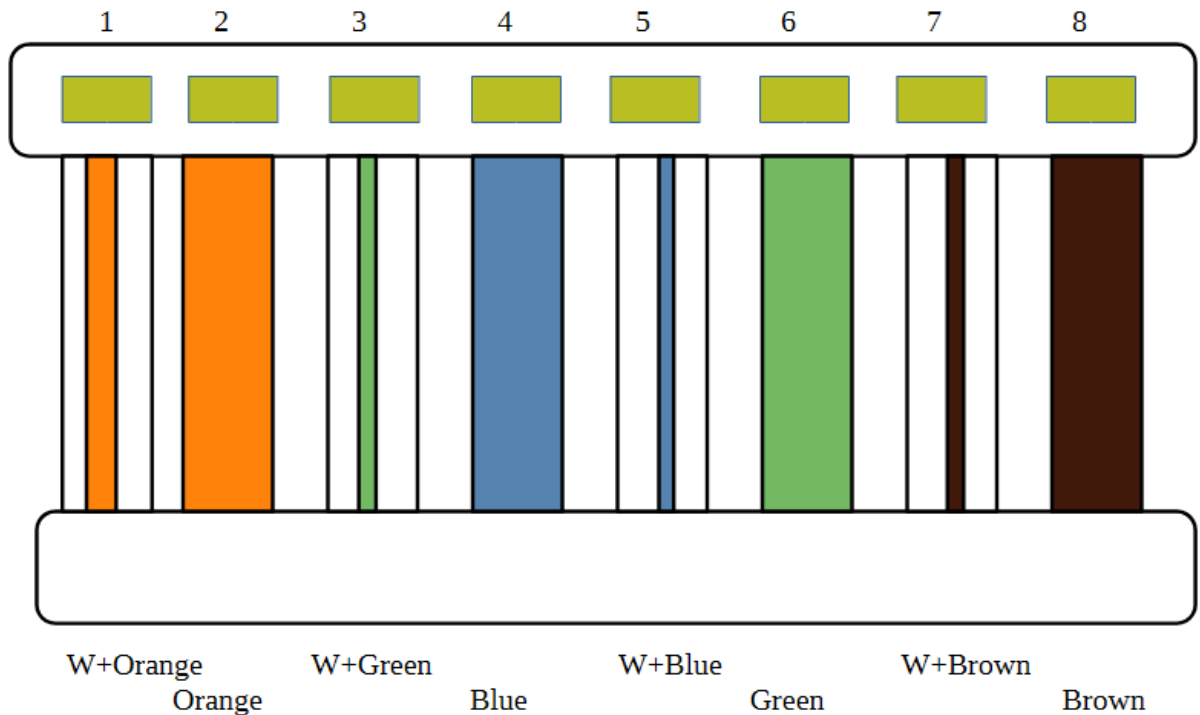


Task 1: Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.



Top row (pins 1–8): the RJ-45 connector housing with 8 gold pin contacts, numbered left to right

Middle section: the 8 individual wires in T568B color order, entering the connector:

1. White/Orange
2. Orange
3. White/Green
4. Blue
5. White/Blue
6. Green
7. White/Brown
8. Brown

Bottom: the cable jacket, showing all 4 twisted pairs bundled inside a single Cat6 sheath.

Task 2: List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

1. Cat6 (Category 6) Ethernet Cable

- **Real-World Application:** Connecting a **Zomato POS (Point of Sale) billing terminal** in a busy restaurant to the local network router.
- **Why it's used:** Cat6 cables support data transfer speeds up to 10 Gbps and feature tight twisting to minimize crosstalk (interference). A commercial restaurant environment is filled with electronic interference from appliances, an ethernet cable carries data in the form of electrical signals, a Cat6 ensures a stable, lag-free connection so order data and payments process instantly without dropping.

2. Fiber Optic Cable

- **Real-World Application:** Connecting a data center hosting an **IPL streaming server** directly to a major Internet Service Provider (ISP) backbone.
- **Why it's used:** Fiber optic cables transmit data as pulses of light rather than electrical signals, allowing for massive bandwidth and almost zero signal degradation over long distances. For a live event like an IPL match with millions of concurrent viewers, fiber optics provide the ultra-high throughput and low latency required to stream 4K video without buffering.

3. Coaxial Cable

- **Real-World Application:** Connecting a residential cable modem/router to the wall outlet provided by a broadband internet provider (like JioFiber or Hathaway coax connections).
- **Why it's used:** Coaxial cables feature a thick copper core surrounded by heavy shielding; they transmit high frequency AC electromagnetic waves. This design is highly effective at carrying high-frequency data signals over moderate distances outdoors and indoors while blocking out external radio frequency interference, making it ideal for delivering internet and cable TV signals into homes.

Task 3: Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

Source: <https://www.youtube.com/watch?v=NWhoJp8UQpo>

Step-by-step: crimping an RJ-45 connector (pass-through method)

1. **Gather your materials:** Ethernet cable (Cat5e, Cat6, or Cat6a depending on your needs), pass-through RJ45 plugs (also called snap plugs), optional strain relief boots, a crimping tool that has a blade for pass-through plugs, wire cutters, a wire stripper, and a cable tester.
2. Cut your cable to length, leaving extra length beyond what you actually need — a few centimeters get cut off during the process.
3. Slide the strain relief boot onto the cable first, it has to go on before you attach the plug.
4. Strip off the outer jacket, being careful not to cut the wires inside.
5. Remove the internal plastic spacer and ripcord thread that run down the center of the cable.
6. Untwist all four pairs into eight individual wires, and straighten it out.
7. Arrange the eight wires in the correct color order to match your chosen standard (T568B: white/orange, orange, white/green, blue, white/blue, green, white/brown, brown), matching whatever standard is already used elsewhere on the network.
8. Straighten and bunch the wire tips evenly, then trim a short, neat length off the ends so they're all level.
9. Feed the wires into the pass-through plug until the outer jacket is fully seated inside — because these plugs are open-ended, you can see through the tip and visually confirm all eight wires are still in the right order before crimping.
10. Crimp the plug — squeezing the tool both presses the contacts into the wires and trims off the excess wire in the same motion.
11. Slide the strain relief boot up over the crimped end.
12. Repeat the same process on the other end, using the same wiring standard.
13. Test the finished cable with a cable tester, which checks all eight connections and confirms the wiring order matches on both ends.

Two common mistakes to avoid

- **Nicking the internal wires while stripping the jacket** — pressing too hard when removing the outer sheath can damage the copper conductors underneath, weakening or breaking a connection.
- **Letting the wire order shift before crimping** — if the grip loosens while arranging the wires, they can swap positions, and with old-style closed plugs this meant snipping off the whole end and starting over.

Task 4: Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

Cat6 UTP cable with RJ-45 connectors for all wired runs between the switch, consoles, and screens (if they have Ethernet-enabled streaming boxes). Cat6 supports Gigabit speeds with low crosstalk, which matters when several devices are streaming or gaming simultaneously and you can't afford lag or buffering during a live match.

A Gigabit network switch as the central hub, since a switch gives every connected device dedicated bandwidth instead of sharing it, which a basic router's ports may not handle well with many devices.

HDMI cables for the actual console-to-screen video connection.

If any screen or console is more than 90–100 meters from the switch it is better to change to **fiber optic with a media converter**, since copper Ethernet degrades past 100 meters while fiber doesn't.

Task 5: Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects. Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.



Type: Fiber Optic cable with an SC/APC connector.

Signal Carried: Light pulses

SC (Standard Connector): Refers to the square, snap-in plastic housing shape of the plug. It utilizes a push-pull latching mechanism to securely click into place.

APC (Angled Physical Contact): The **green color** is a universal industry standard indicating that the fiber tip inside is polished at an 8 deg. angle. This angled tip ensures that any internal light reflections bounce outward into the cable cladding rather than traveling straight back down the core, resulting in much better signal quality.